Strategically mitigating the impacts of offshore wind on marine fisheries can also promote climate resilience

Abstract

Offshore wind (OSW) is expanding as a source of renewable energy but is expected to displace and negatively impact marine fisheries. While tools for avoiding and minimizing these conflicts through marine spatial planning are well-established, there is little guidance on mitigating the unavoided impacts. We reviewed current and proposed fisheries mitigation plans to develop a comprehensive menu of strategies for offsetting the socioeconomic impacts of OSW development and assessed the extent to which these strategies also build long-term climate resilience. Our review identified 60 mitigation measures grouped within 25 broader mitigation strategies. The identified strategies extended beyond direct financial compensation to include investments in port infrastructure, seafood marketing, workforce development, fisheries science, governance participation, vessel safety, and other initiatives. The identified mitigation strategies offer strong support for socioeconomic resilience and moderate support for resilience in governance but offer weak support for ecological resilience. Most of the identified strategies could be financed, implemented, and enforced through community benefit agreements (contractual agreements between OSW project developers and impacted communities) in close partnership with local business, regulatory, educational, and research institutions. Although all of the identified mitigation strategies represent fisheries subsidies, many of which are capacity-enhancing, most are not harmful when governance is strong. We envision that this menu of mitigation strategies could be used to strategically design mitigation plans that diversify their support for long-term resilience when offsetting the impacts of OSW development and other emerging ocean uses (e.g., aquaculture, oil/gas/mineral extraction, protection), on marine fisheries.

Keywords: wind energy; mitigation; compensation; avoidance; minimization; community benefit agreements

Highlights

1. Offshore wind is a growing source of green energy but can negatively impact fisheries
2. We reviewed mitigation strategies and their support of long-term climate resilience
3. The identified mitigation strategies strongly support socioeconomic resilience
4. Governance and ecological resilience are moderately and poorly supported, respectively
5. These strategies can also be used to mitigate the impacts of other emerging ocean uses